Research Methodology

Week 8 Assignment: Questions, Answers and Explanations

Week 8: Department Specific Discussions

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

Correct answers are the NPTEL portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer. This week mixes research design, dissertation structure and general knowledge. You scored full marks on every question.

Answer key at a glance

Question	Type	Correct option(s)	Your portal score
1	Single answer	**B**	1 (full marks)
2	Single answer	**D**	1 (full marks)
3	Single answer	**C**	1 (full marks)
4	Single answer	**B**	1 (full marks)
5	Single answer	**A**	1 (full marks)
6	Single answer	**D**	1 (full marks)
7	Single answer	**D**	1 (full marks)
8	Single answer	**D**	1 (full marks)
9	Single answer	**C**	1 (full marks)
10	Single answer	**A**	1 (full marks)

Questions with explanations

Question 1 of 10

In order to pursue the research, which of the following is priorly required?

Single answer.

Option	Statement	Verdict	Reason
A	Developing a research design	**Wrong**	The design follows once you know the question.
B	**Formulating a research question**	**Correct**	The question says what you want to find out; everything else follows from it.
C	Deciding about the data analysis procedure	**Wrong**	The analysis plan depends on the question and design.
D	Formulating a research hypothesis	**Wrong**	A hypothesis is a tentative answer to the question, so it comes after it.

Correct answer: B.

Explanation: Research starts with a question. The hypothesis, the design and the analysis plan are all built to answer it.

Question 2 of 10

Which one among the following statements is false in the context of participatory research?

Single answer.

Option	Statement	Verdict	Reason
A	It recognizes knowledge as power	**Wrong**	True: participatory research aims to empower people through knowledge.
B	It is a collective process of inquiry	**Wrong**	True: the community takes part in the inquiry.
C	It emphasizes people as experts	**Wrong**	True: local people are treated as experts on their own situation.
D	**Its sole purpose is the production of knowledge**	**Correct**	False: it also aims at empowerment and action for change, not only knowledge.

Correct answer: D.

Explanation: Participatory research involves the people studied as co-researchers and aims to empower them and bring change. Saying its sole purpose is to produce knowledge is false.

Note: *This question asks for the FALSE statement, so "Correct" marks the false statement you should select.*

Question 3 of 10

How to judge the depth of any research?

Single answer.

Option	Statement	Verdict	Reason
A	By research title	**Wrong**	A title can be impressive and empty.
B	By research duration	**Wrong**	Long duration does not mean deep research.
C	**By research objectives**	**Correct**	The objectives show how deep the questions go and what the work sets out to achieve.
D	By total expenditure on research	**Wrong**	Cost says nothing about depth.

Correct answer: C.

Explanation: The objectives state what the research tries to find out, so they show how deep it is. Title, duration and cost are only surface features.

Question 4 of 10

What is the main aim of interdisciplinary research?

Single answer.

Option	Statement	Verdict	Reason
A	To over simplify the problem of research	**Wrong**	It aims to solve complex problems, not oversimplify them.
B	**Integrating knowledge from multiple disciplines to solve a problem.**	**Correct**	It combines the methods and knowledge of several fields to solve one problem.
C	To create a new trend in research methodology	**Wrong**	A new trend may result, but it is not the aim.
D	To reduce the emphasis on a single subject in the research domain	**Wrong**	It is a by-product, not the main aim.

Correct answer: B.

Explanation: Some problems cannot be solved by one field alone. Interdisciplinary research combines knowledge from several disciplines to solve such a problem.

Question 5 of 10

Electron microscopes can be used in reverse to write in extremely small areas

Single answer.

Option	Statement	Verdict	Reason
A	**True**	**Correct**	The focused electron beam of a microscope can expose a resist and draw patterns a few nanometres wide (electron-beam lithography).
B	False	**Wrong**	Electron-beam writing is a real technique.

Correct answer: A.

Explanation: An electron microscope focuses a very fine electron beam to look at a sample. Used in reverse, the same beam can write extremely small patterns, which is called electron-beam lithography.

Note: *This fact comes from the department-specific talks and general science, not from a calculation, so remember it as a fact.*

Question 6 of 10

What are the core elements of a dissertation?

Single answer.

Option	Statement	Verdict	Reason
A	Introduction; Data Collection; Data Analysis; Conclusions and Recommendations	**Wrong**	It leaves out the literature review and methodology.
B	Executive Summary; Literature Review; Data Gathered; Conclusions; Bibliography	**Wrong**	The executive summary suits business reports; methodology is missing.
C	Research Plan; Research Data; Analysis; References	**Wrong**	It has no introduction, literature review or conclusions.
D	**Introduction; Literature Review; Research Methodology; Results; Discussions and Conclusions**	**Correct**	This is the standard order of a dissertation.

Correct answer: D.

Explanation: A dissertation introduces the problem, reviews what is known, explains the method, presents the results and then discusses them and concludes.

Question 7 of 10

How can we enhance the research objective?

Single answer.

Option	Statement	Verdict	Reason
A	By making it more valid	**Wrong**	True, but only one of the ways.
B	By making it more reliable	**Wrong**	True, but only one of the ways.
C	By making it more impartial	**Wrong**	True, but only one of the ways.
D	**All of the above**	**Correct**	Validity, reliability and impartiality each strengthen an objective.

Correct answer: D.

Explanation: An objective is stronger when it measures what it claims to (valid), gives consistent results (reliable) and is free of the researcher's bias (impartial).

Question 8 of 10

A common test in research demands much priority on

Single answer.

Option	Statement	Verdict	Reason
A	Reliability	**Wrong**	True, but only one of the criteria.
B	Useability	**Wrong**	True, but only one of the criteria.
C	Objectivity	**Wrong**	True, but only one of the criteria.
D	**All of the above**	**Correct**	A good test needs all three.

Correct answer: D.

Explanation: A good test gives consistent results (reliability), is practical to use (usability) and gives the same result whoever marks it (objectivity).

Question 9 of 10

An appropriate source to find out descriptive information is

Single answer.

Option	Statement	Verdict	Reason
A	Bibliography	**Wrong**	A bibliography lists publications.
B	Directory	**Wrong**	A directory lists names and addresses.
C	**Encyclopedia**	**Correct**	An encyclopedia gives descriptive, explanatory information on a topic.
D	Dictionary	**Wrong**	A dictionary gives meanings of words.

Correct answer: C.

Explanation: For a description of a topic, an encyclopedia is the right reference source.

Question 10 of 10

Mahabalipuram in south-east India, pre-eminently testimony to the ..... civilization

Single answer.

Option	Statement	Verdict	Reason
A	**Pallavas**	**Correct**	The monuments at Mahabalipuram were built by the Pallava dynasty in the 7th and 8th centuries.
B	Mauryas	**Wrong**	The Mauryan period (about the 3rd century BCE) is many centuries earlier than these monuments.
C	Guptas	**Wrong**	The Guptas ruled in the north.
D	Cholas	**Wrong**	The Cholas are known for temples such as Thanjavur, not Mahabalipuram.

Correct answer: A.

Explanation: Mahabalipuram's rock-cut temples, the Shore Temple and carved reliefs belong to the Pallava dynasty.

Note: *This is a general-knowledge fact that the lectures do not teach, so remember it as a fact.*